

Fluxo Máximo (1)

Zenilton Patrocínio

Rede de Fluxo

Uma rede de fluxo é um grafo direcionado e ponderado $G = (V, E)$ em que se associa a cada aresta $e \in E$ um valor de capacidade $u(e) > 0$.

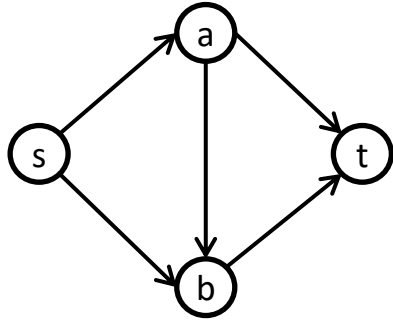
Existem dois vértices especiais em uma rede de fluxo:

- **Vértice s :** “*source*” (ou fonte) que representa a origem fluxo; e
- **Vértice t :** “*terminal*” (ou sumidouro) que representa o destino do fluxo.

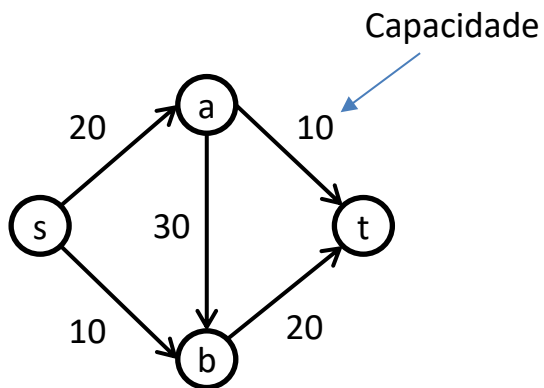
Os demais nós da rede são denominados nós internos.

Assume-se que não há arestas entrando em s nem saindo de t , que todo vértice possui pelo menos uma aresta incidente a ele e que as capacidades são inteiras.

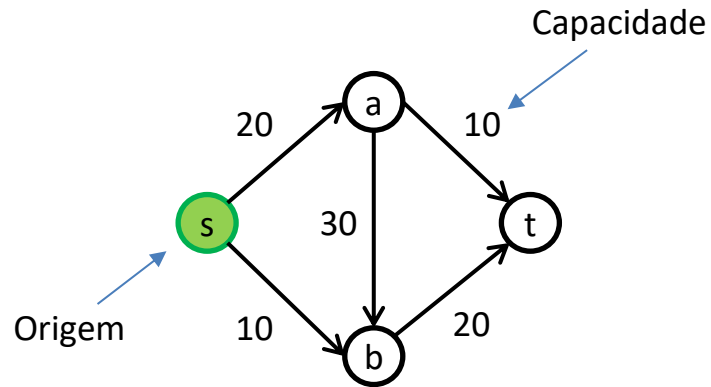
Rede de Fluxo – Exemplo



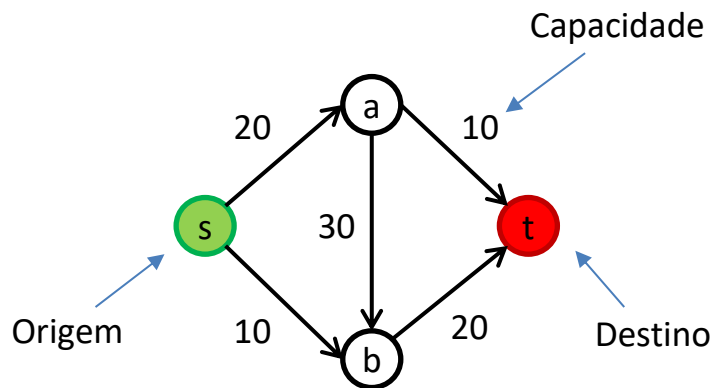
Rede de Fluxo – Exemplo



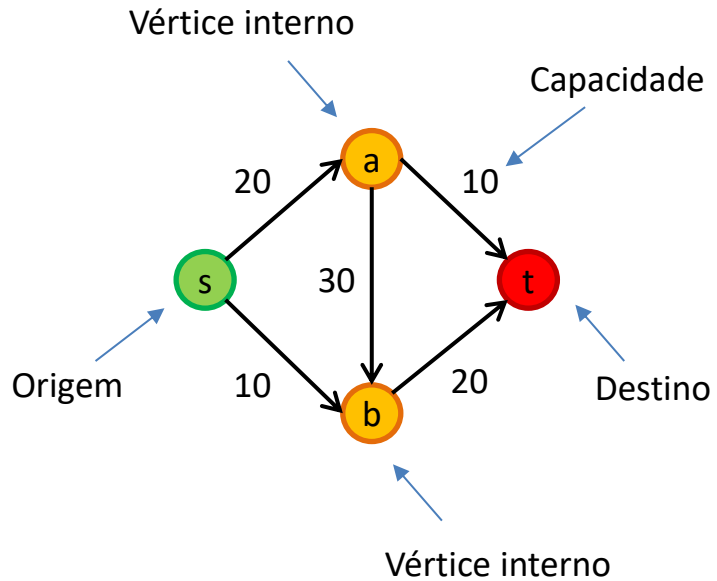
Rede de Fluxo – Exemplo



Rede de Fluxo – Exemplo



Rede de Fluxo – Exemplo



Fluxo

Um fluxo f de s a t em uma rede é uma função que associa a cada aresta $e \in E$ um número real não negativo $f(e)$ satisfazendo às seguintes condições:

- **Condição de capacidade:** Para toda aresta $e \in E$, seu valor de fluxo é não negativo e não pode exercer sua capacidade, isto é,

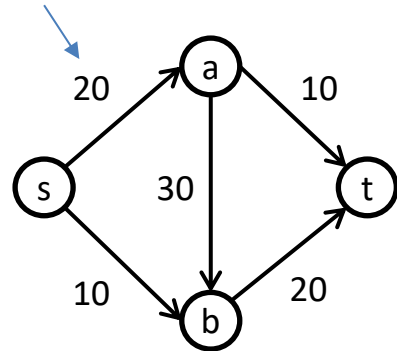
$$0 \leq f(e) \leq u(e)$$

- **Condição de conservação:** Para todo vértice interno v , a soma dos fluxos das arestas que entram em v é igual ao total de fluxo das arestas que saem de v , isto é,

$$\sum_{e \in \Gamma^-(v)} f(e) = \sum_{e \in \Gamma^+(v)} f(e)$$

Fluxo – Exemplo

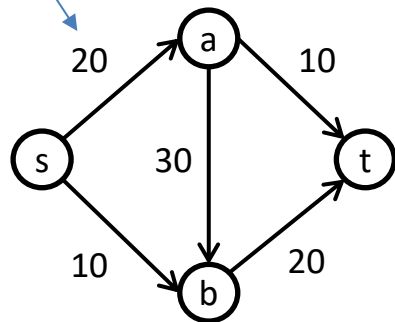
Capacidade



Rede de Fluxo

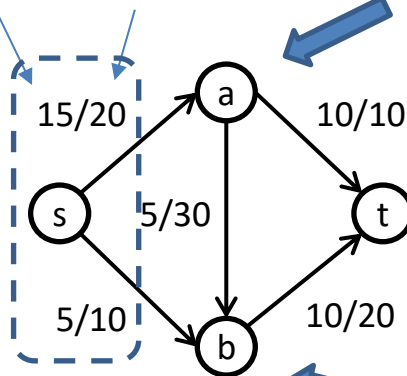
Fluxo – Exemplo

Capacidade



Rede de Fluxo

Fluxo / Capacidade



Fluxo = 20

Fluxo Viável

$$\sum_{e \in \Gamma^-(a)} f(e) = \sum_{e \in \Gamma^+(a)} f(e)$$

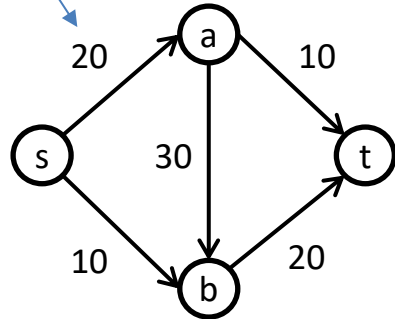


Valor total do fluxo
=
Fluxo que sai da origem

$$\sum_{e \in \Gamma^-(b)} f(e) = \sum_{e \in \Gamma^+(b)} f(e)$$

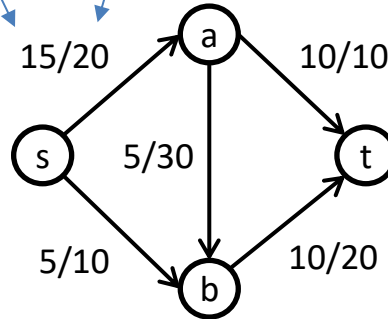
Fluxo – Exemplo

Capacidade



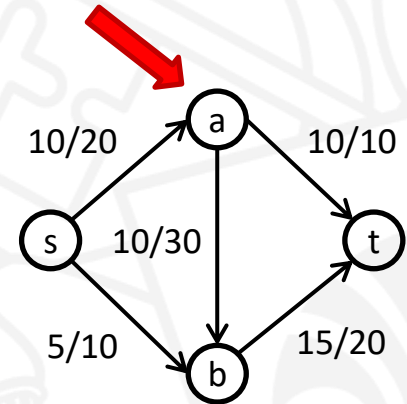
Rede de Fluxo

Fluxo / Capacidade



Fluxo Viável

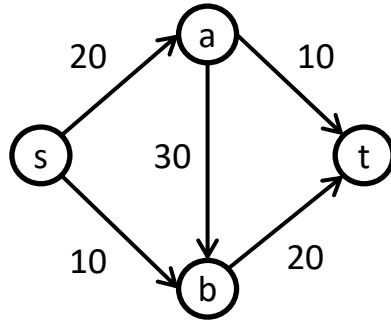
$$\sum_{e \in \Gamma^-(a)} f(e) \neq \sum_{e \in \Gamma^+(a)} f(e)$$



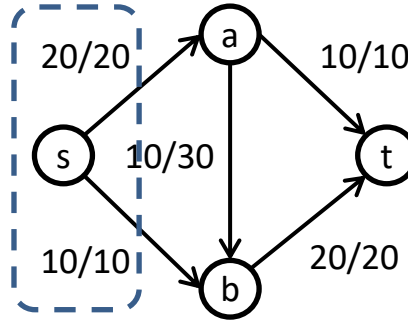
Fluxo Inviável

Fluxo Máximo

Dada uma rede, o problema de fluxo máximo consiste em determinar o maior valor de fluxo viável entre a fonte **s** e o sumidouro **t**.



Rede de Fluxo



Fluxo Máximo



Fluxo Máximo = 30

Método de Ford-Fulkerson

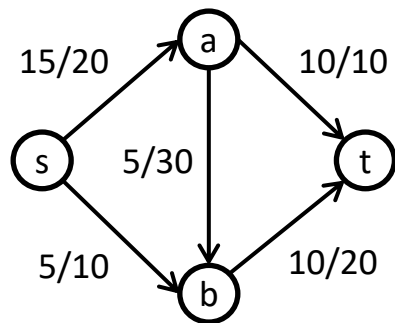
Rede Residual

Dado um fluxo f em uma rede $G = (V, E)$, a **rede residual** $G'(f)$ é um grafo direcionado ponderado que:

- Possui os mesmos vértices que G , isto é, $V(G') = V(G)$;
- Para toda aresta $e = (v, w) \in E$ tal que $f(e) < u(e)$, $G'(f)$ contém a **aresta direta (v, w)** com capacidade (residual) igual a **$u_r(e) = u(e) - f(e)$** ;
- Para toda aresta $e = (v, w) \in E$ tal que $f(e) > 0$, $G'(f)$ contém a **aresta reversa (w, v)** com capacidade (residual) igual a **$u_r(e) = f(e)$** .

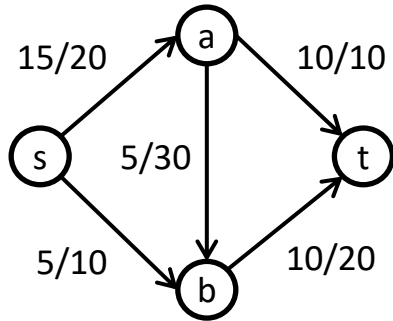
Um caminho na rede residual saindo da fonte s até o sumidouro t é chamado de **caminho aumentante** (ou caminho de aumento de fluxo).

Rede Residual – Exemplo

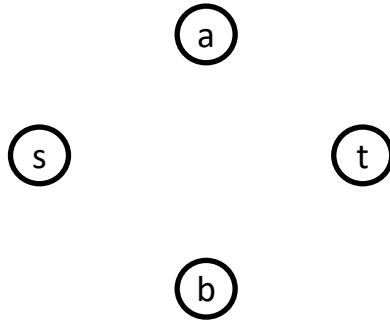


Fluxo Viável

Rede Residual – Exemplo

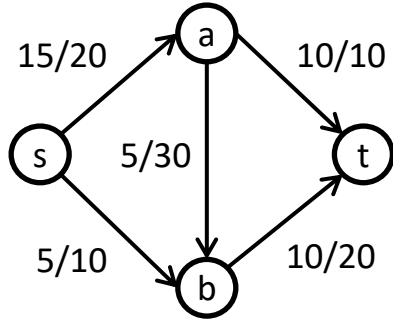


Fluxo Viável

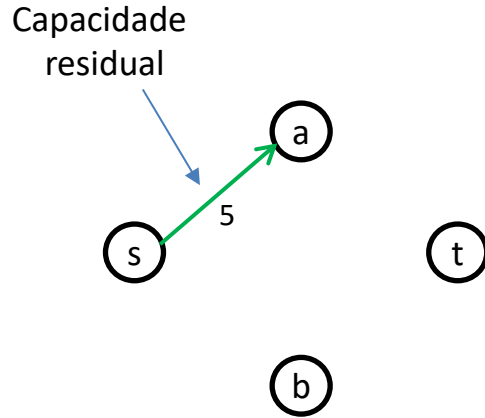


Rede Residual

Rede Residual – Exemplo

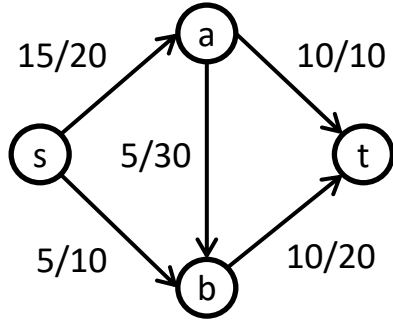


Fluxo Viável

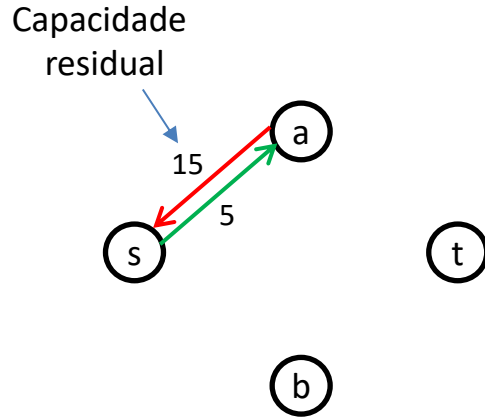


Rede Residual

Rede Residual – Exemplo

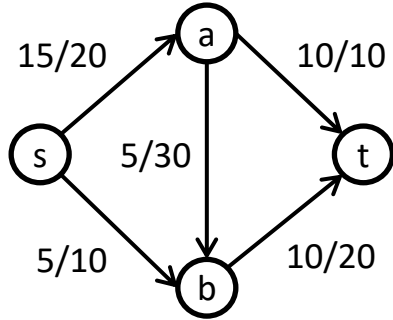


Fluxo Viável

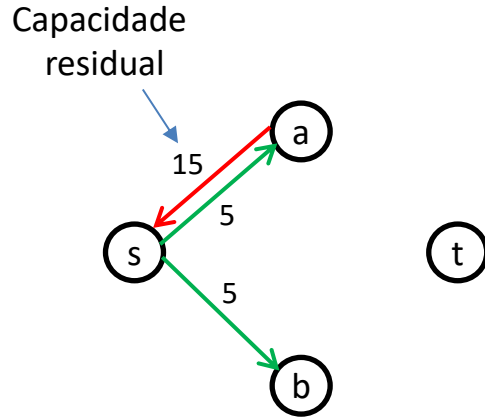


Rede Residual

Rede Residual – Exemplo

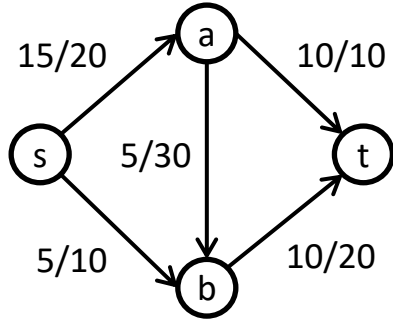


Fluxo Viável

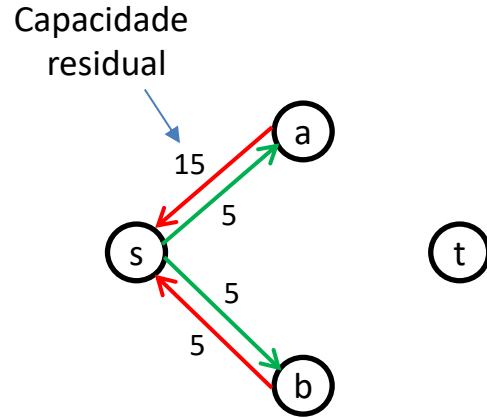


Rede Residual

Rede Residual – Exemplo

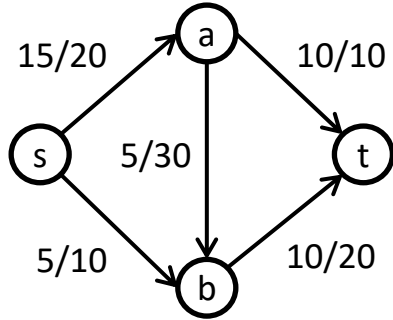


Fluxo Viável

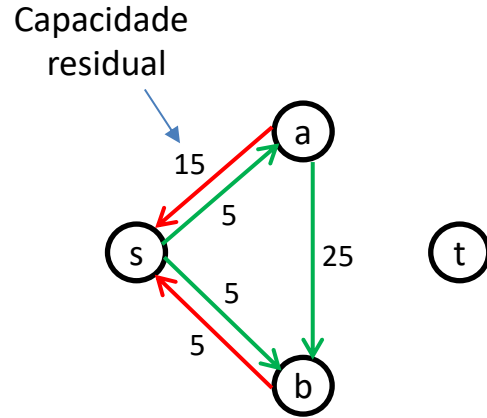


Rede Residual

Rede Residual – Exemplo

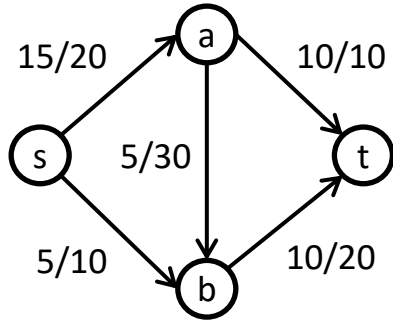


Fluxo Viável

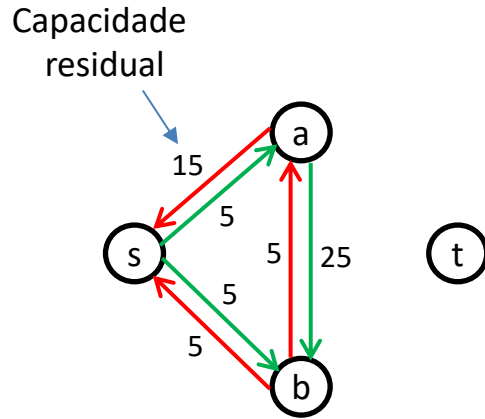


Rede Residual

Rede Residual – Exemplo

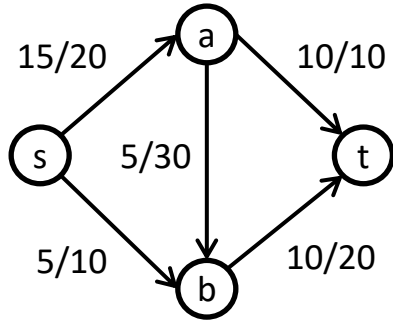


Fluxo Viável



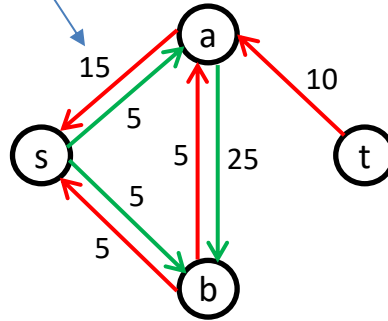
Rede Residual

Rede Residual – Exemplo



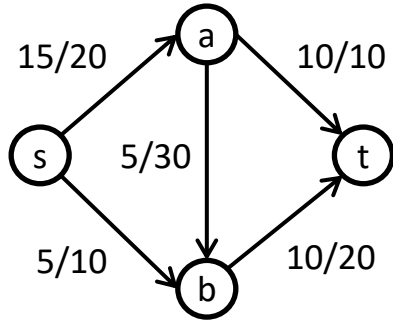
Fluxo Viável

Capacidade residual



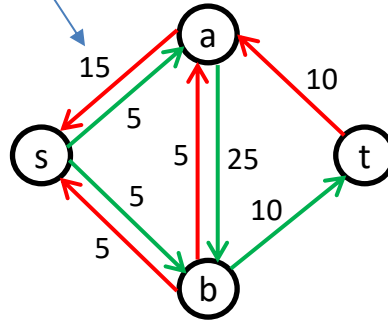
Rede Residual

Rede Residual – Exemplo



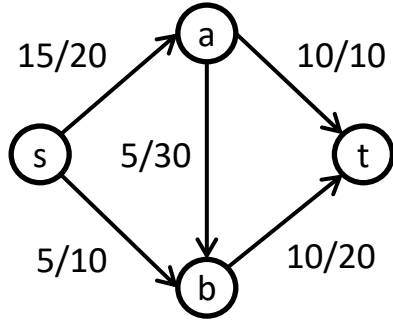
Fluxo Viável

Capacidade residual



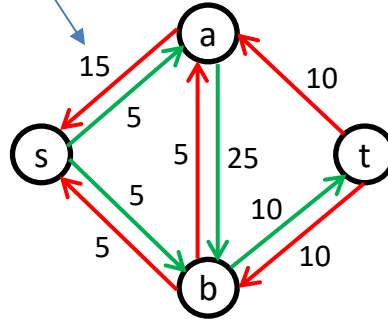
Rede Residual

Rede Residual – Exemplo



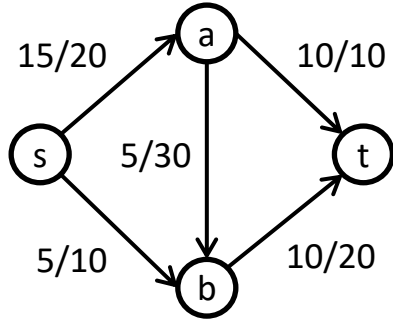
Fluxo Viável

Capacidade residual



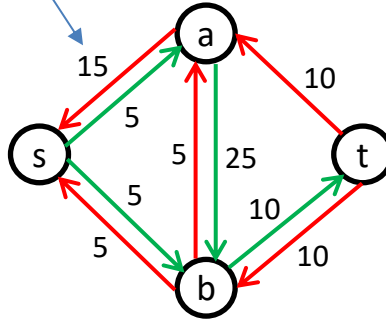
Rede Residual

Rede Residual – Exemplo

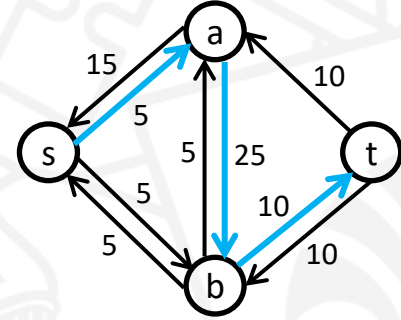


Fluxo Viável

Capacidade residual

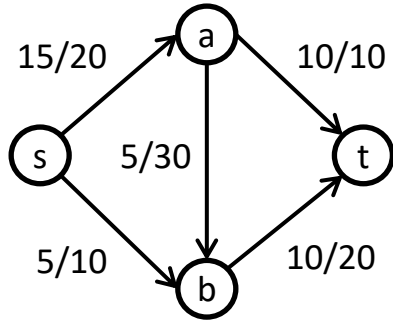


Rede Residual



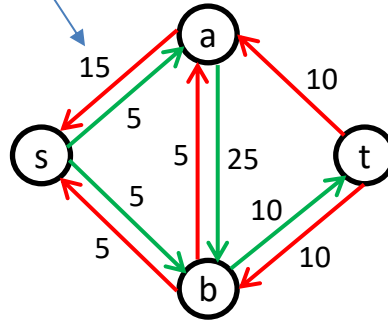
Caminho Aumentante

Rede Residual – Exemplo



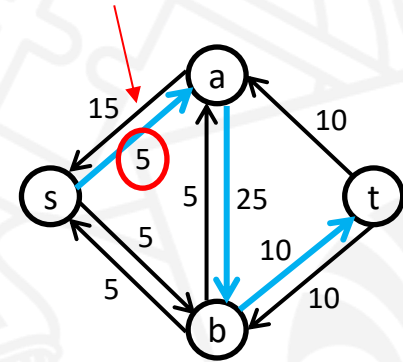
Fluxo Viável

Capacidade residual



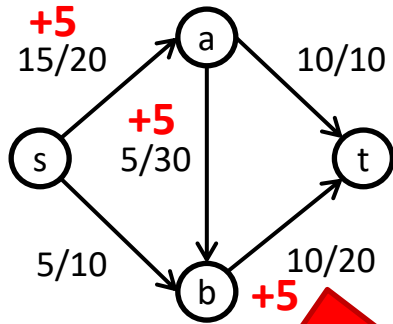
Rede Residual

Menor capacidade do caminho aumentante – “Gargalo” ou Δ



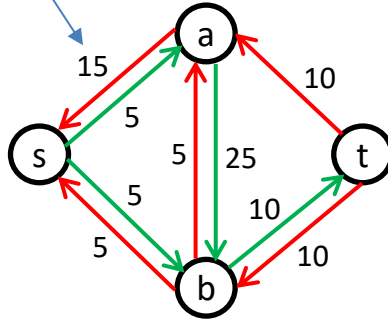
Caminho Aumentante

Rede Residual – Exemplo



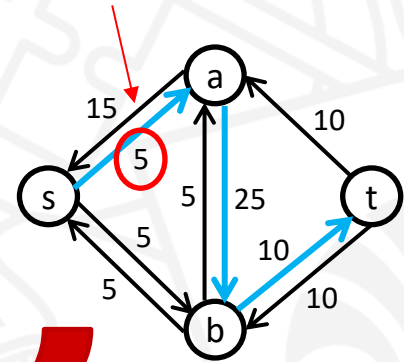
Fluxo Viável

Capacidade residual



Rede Residual

Menor capacidade do caminho
aumentante – “Gargalo” ou Δ

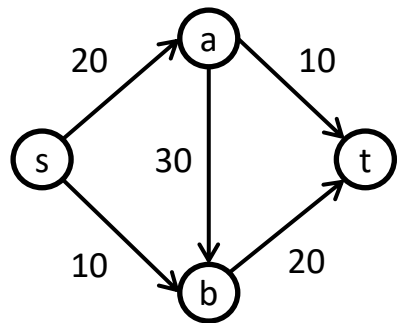


Caminho Aumentante

Método Ford-Fulkerson – Algoritmo

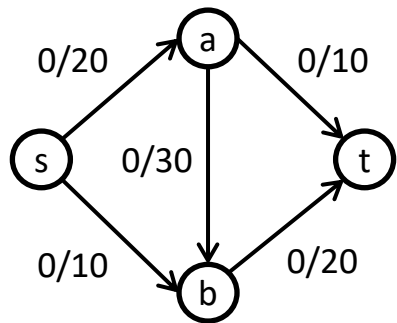
1. para toda aresta $e \in E(G)$ faça $f(e) \leftarrow 0$; // Inicializar fluxo
2. Construir a rede residual $G'(f)$ // Construir rede residual inicial
3. enquanto existir algum caminho aumentante P em $G'(f)$ efetuar
 - a. $\Delta = \min \{ u_r(e) \mid e \in P \}$; // Determinar “gargalo” de P
 - b. para cada aresta $(v, w) \in P$ faça
 - i. se (v, w) for aresta direta então
 $f(v, w) \leftarrow f(v, w) + \Delta$ // Aumentar fluxo
 - ii. senão
 $f(w, v) \leftarrow f(w, v) - \Delta$ // Reduzir fluxo
 - c. Atualizar a rede residual $G'(f)$ // Construir nova rede residual

Método Ford-Fulkerson – Exemplo



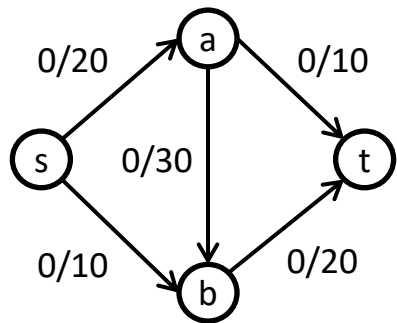
Rede de Fluxo

Método Ford-Fulkerson – Exemplo

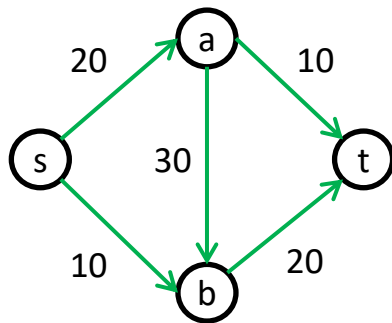


Fluxo Viável

Método Ford-Fulkerson – Exemplo

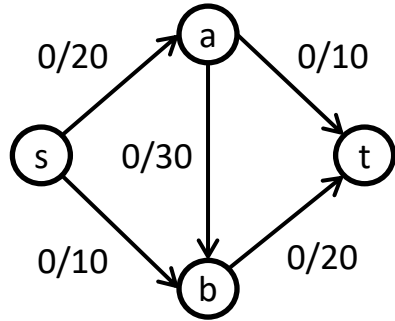


Fluxo Viável

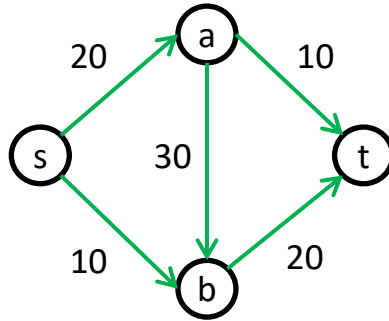


Rede Residual

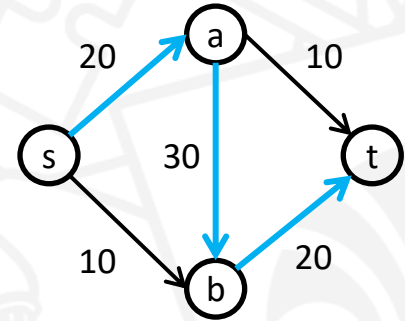
Método Ford-Fulkerson – Exemplo



Fluxo Viável

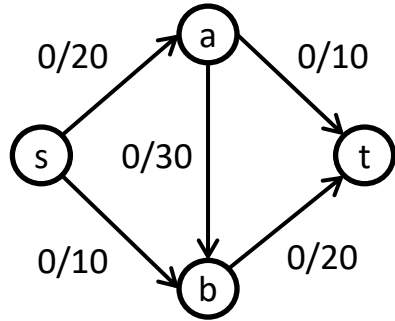


Rede Residual

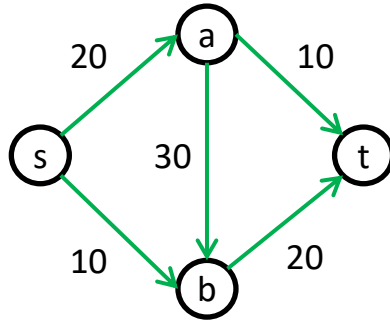


Caminho Aumentante

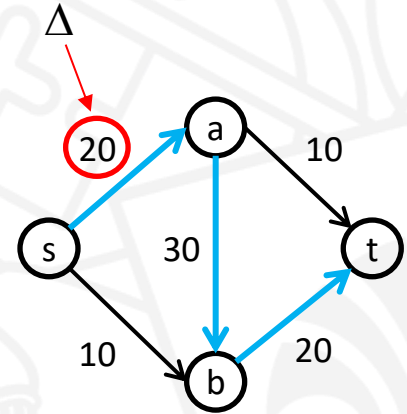
Método Ford-Fulkerson – Exemplo



Fluxo Viável

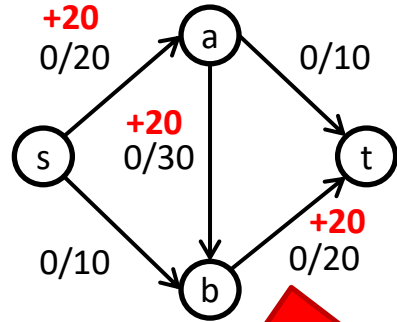


Rede Residual

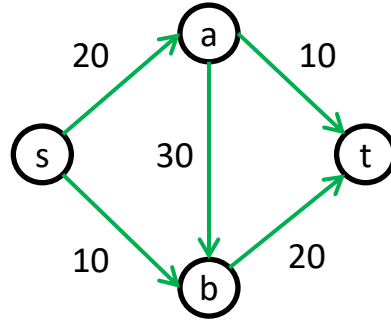


Caminho Aumentante

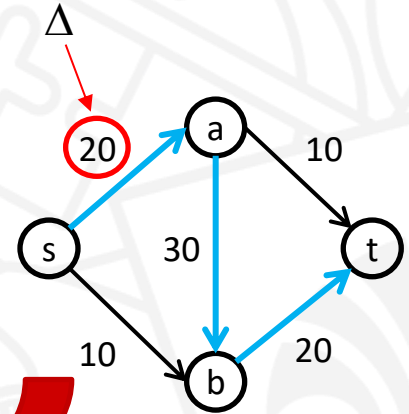
Método Ford-Fulkerson – Exemplo



Fluxo Viável

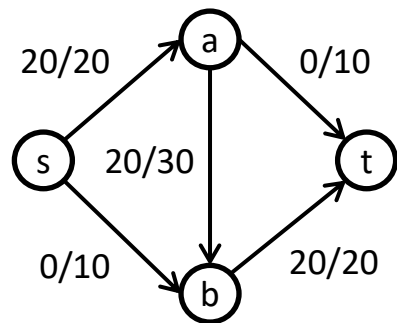


Rede Residual



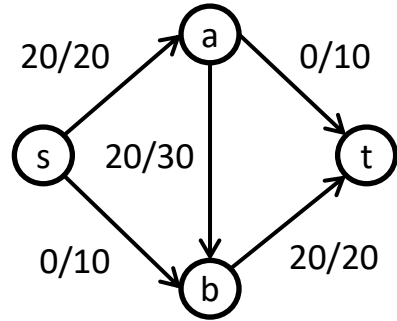
Caminho Aumentante

Método Ford-Fulkerson – Exemplo

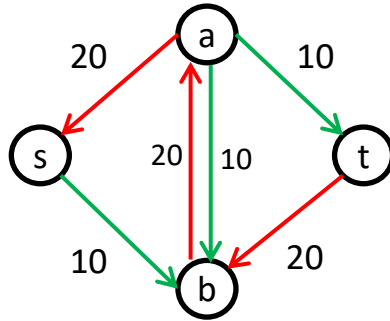


Fluxo Viável

Método Ford-Fulkerson – Exemplo

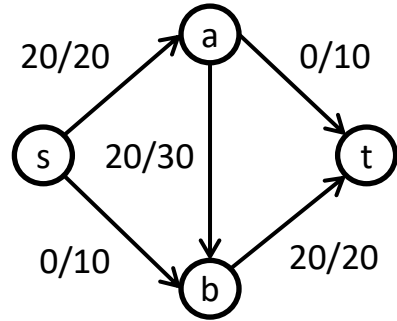


Fluxo Viável

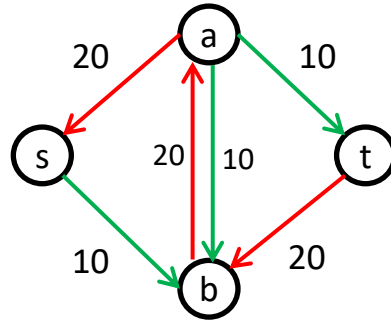


Rede Residual

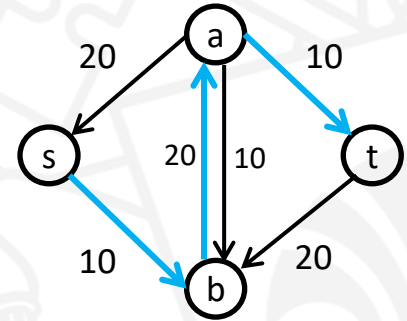
Método Ford-Fulkerson – Exemplo



Fluxo Viável

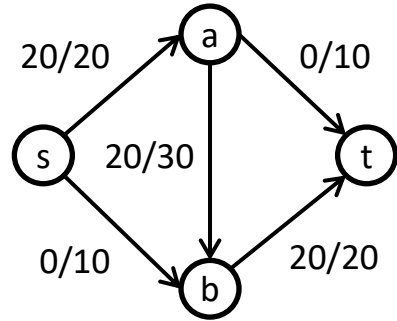


Rede Residual

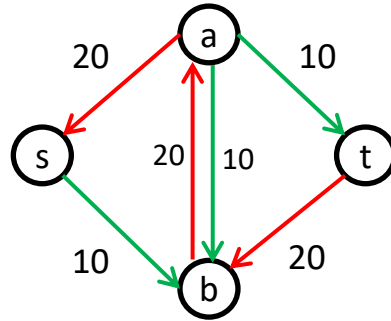


Caminho Aumentante

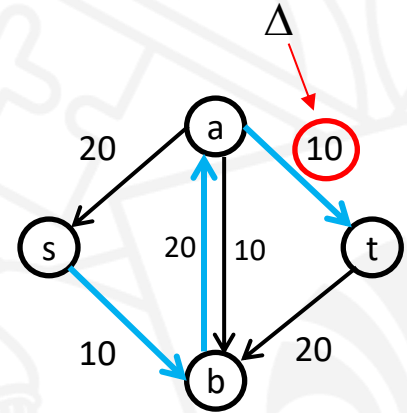
Método Ford-Fulkerson – Exemplo



Fluxo Viável

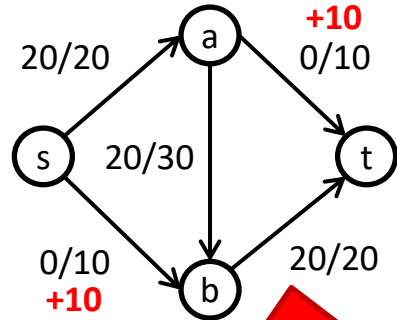


Rede Residual

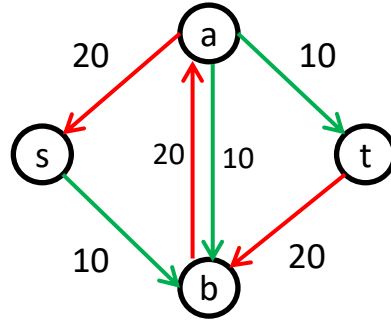


Caminho Aumentante

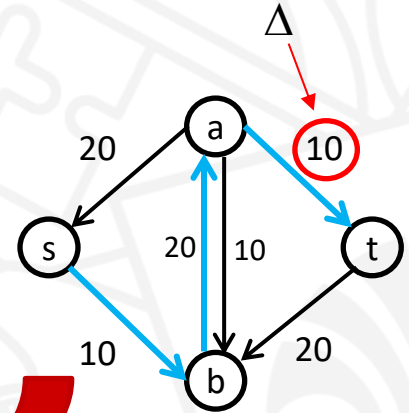
Método Ford-Fulkerson – Exemplo



Fluxo Viável

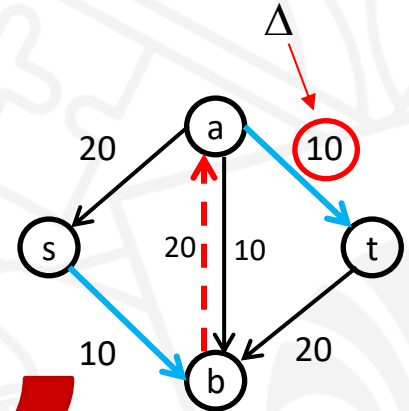
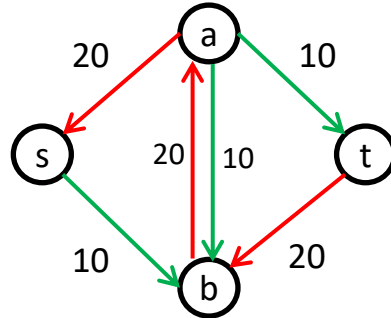
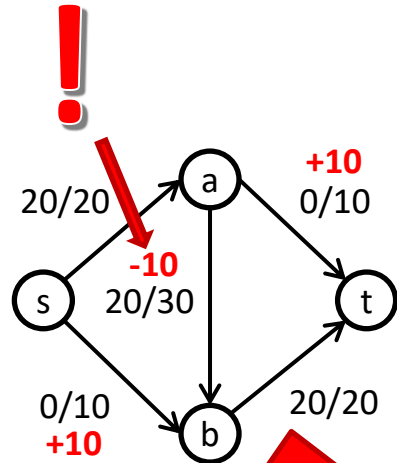


Rede Residual

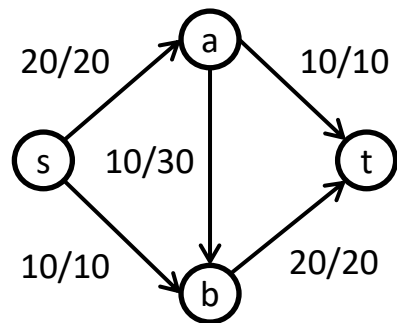


Caminho Aumentante

Método Ford-Fulkerson – Exemplo

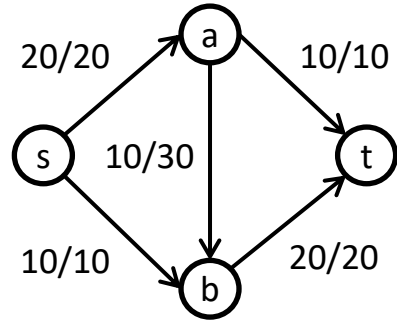


Método Ford-Fulkerson – Exemplo

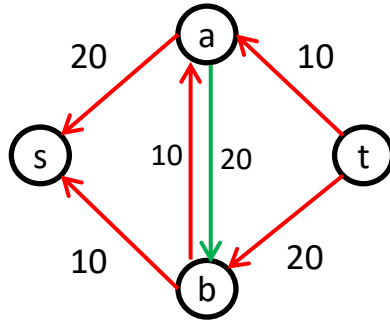


Fluxo Viável

Método Ford-Fulkerson – Exemplo

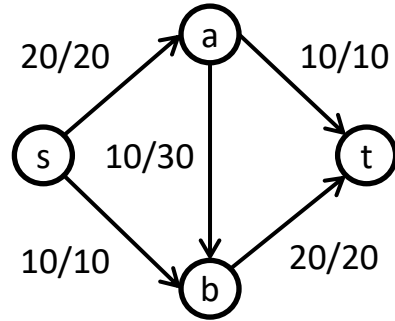


Fluxo Viável

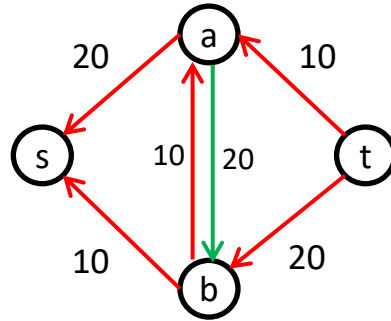


Rede Residual

Método Ford-Fulkerson – Exemplo



Fluxo Viável

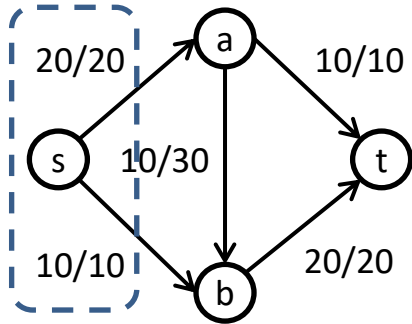


Rede Residual

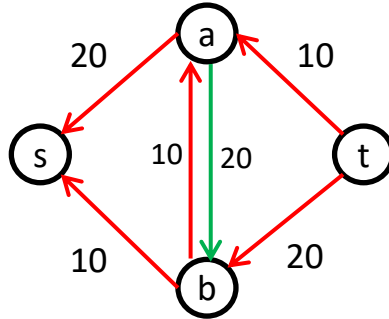


Não existe caminho
aumentante

Método Ford-Fulkerson – Exemplo



Fluxo Viável



Rede Residual



Não existe caminho
aumentante

Fluxo máximo = 30

Teorema do Fluxo Máximo e Corte Mínimo

Teorema do Fluxo Máximo e Corte Mínimo

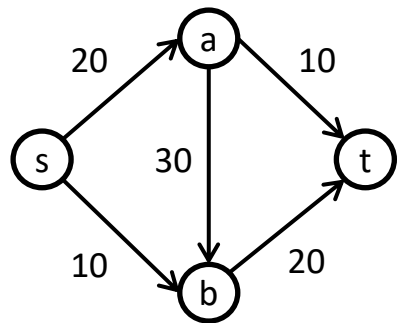
Dada uma rede $G = (V, E)$ e um subconjunto $S \subset V$, tal que a fonte $s \in S$ e o sumidouro $t \notin S$.

O corte(S) – chamado de corte s - t da rede de fluxo – contém as arestas (v, w) em que vértice $v \in S$ e o vértice $w \notin S$.

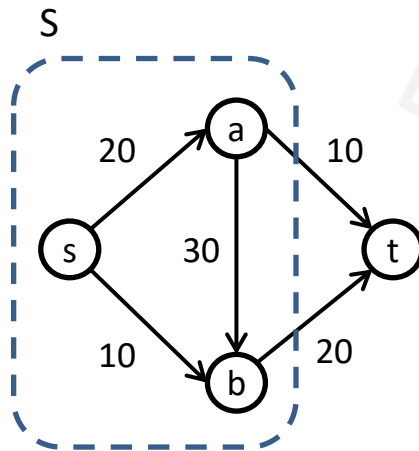
A capacidade do corte(S) é dada pela soma das capacidades de suas arestas.

TEOREMA: Em qualquer rede de fluxo, o valor do fluxo máximo entre a fonte s e o sumidouro t é igual à capacidade do corte s - t mínimo da rede.

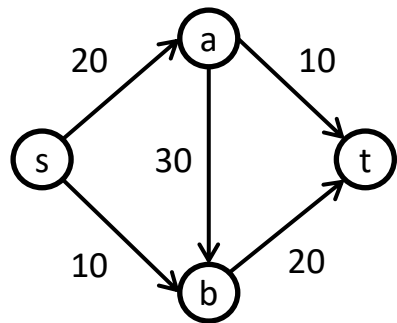
Teorema do Fluxo Máximo e Corte Mínimo



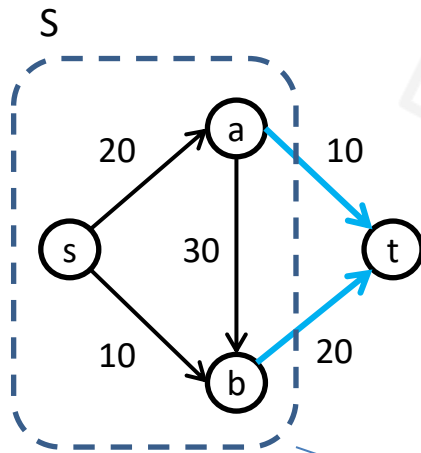
Rede de Fluxo



Teorema do Fluxo Máximo e Corte Mínimo

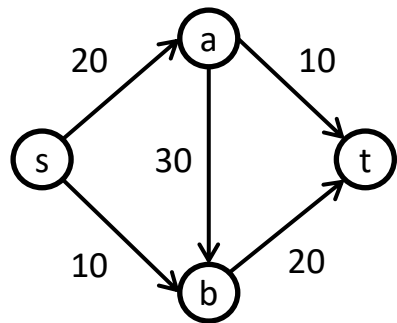


Rede de Fluxo

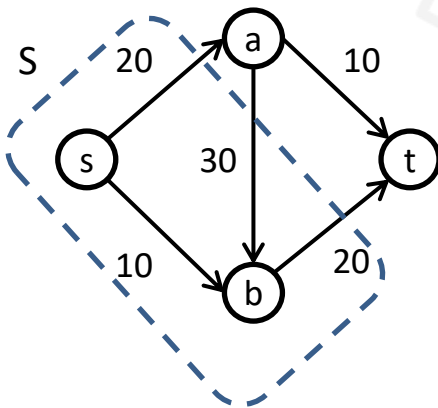


Capacidade corte(S) = 30

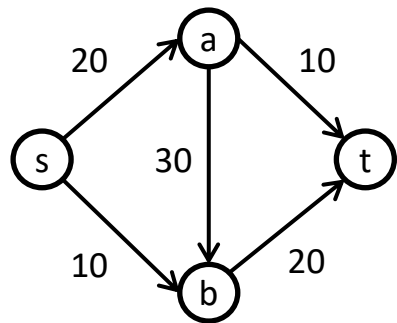
Teorema do Fluxo Máximo e Corte Mínimo



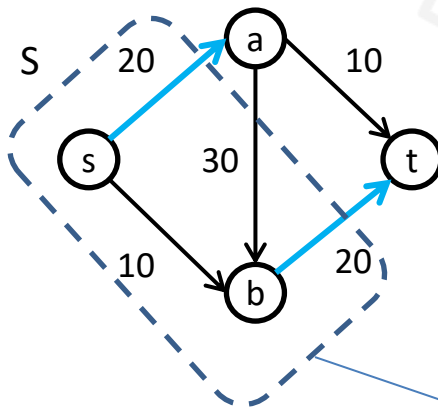
Rede de Fluxo



Teorema do Fluxo Máximo e Corte Mínimo

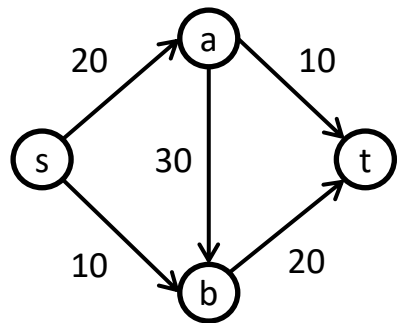


Rede de Fluxo

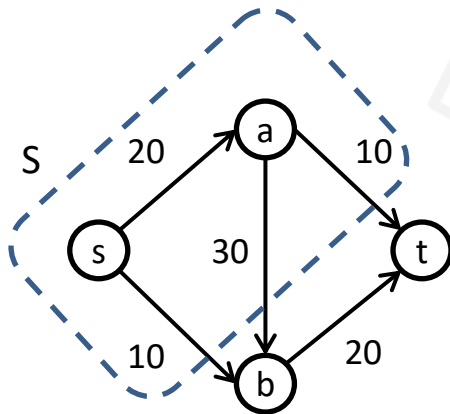


Capacidade corte(S) = 40

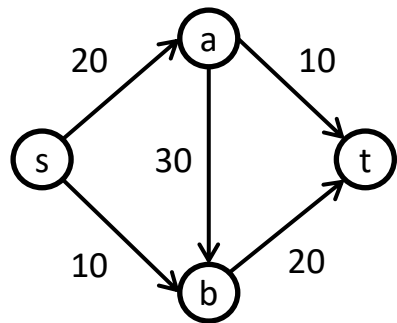
Teorema do Fluxo Máximo e Corte Mínimo



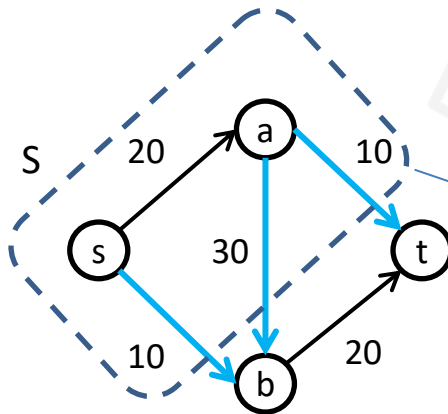
Rede de Fluxo



Teorema do Fluxo Máximo e Corte Mínimo

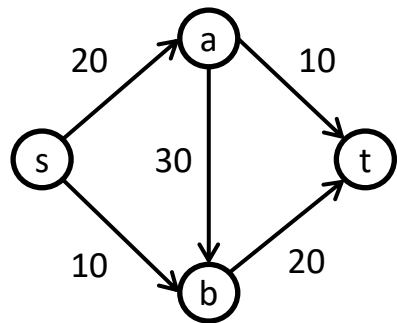


Rede de Fluxo

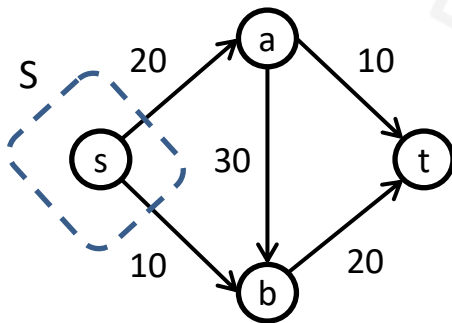


Capacidade corte(S) = 50

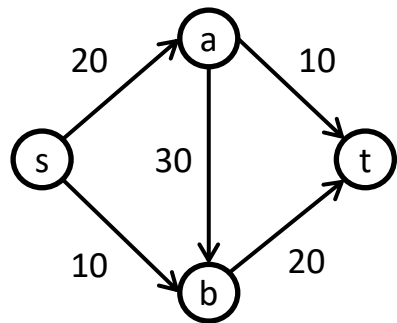
Teorema do Fluxo Máximo e Corte Mínimo



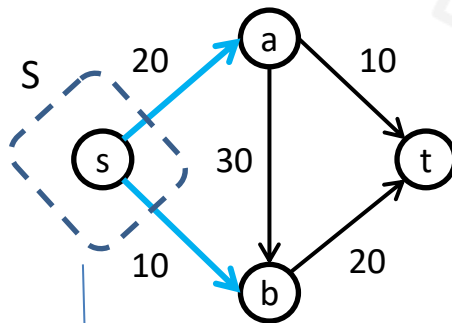
Rede de Fluxo



Teorema do Fluxo Máximo e Corte Mínimo

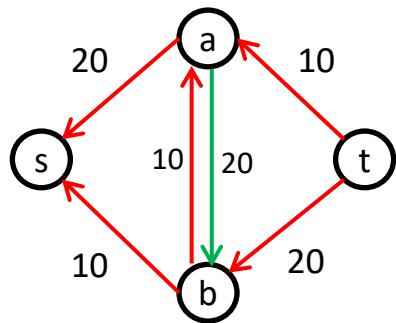


Rede de Fluxo



Capacidade corte(S) = 30

Teorema do Fluxo Máximo e Corte Mínimo

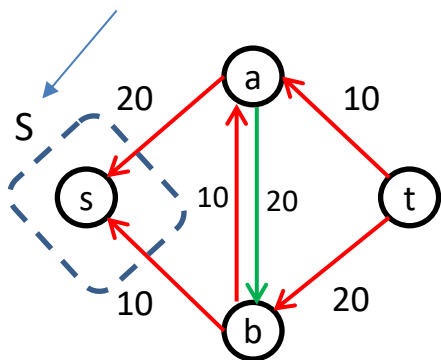


Rede Residual

Fluxo máximo = 30

Teorema do Fluxo Máximo e Corte Mínimo

Na solução ótima, S é o conjunto dos elementos alcançáveis a partir da fonte s

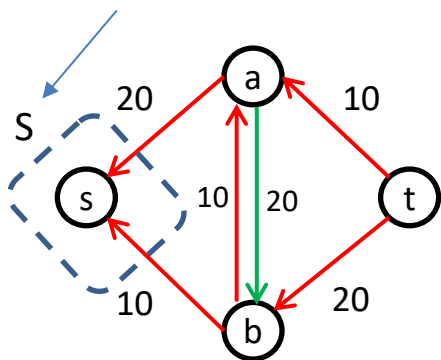


Rede Residual

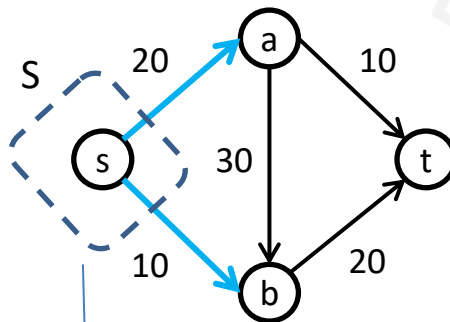
Fluxo máximo = 30

Teorema do Fluxo Máximo e Corte Mínimo

Na solução ótima, S é o conjunto dos elementos alcançáveis a partir da fonte s



Rede Residual



Capacidade corte(S) = 30

Fluxo máximo = 30

